

Relation in Neo4j only 1 label.

Instructions: Log into Neo4j (using the Sandbox (<https://neo4j.com/sandbox/>)) and create an instance of the *Recommendations* database.

Additional resources: For additional help with Neo4j please feel free to explore the material in the free course, Neo4j Graph Academy (<https://graphacademy.neo4j.com/>), especially the *Beginner* course modules. Also see the Cypher documentation (<https://neo4j.com/docs/cypher-manual/current/introduction/>) for further help.

1. Our first Cypher queries in Neo4j: Execute the following queries and write down the result. Do you notice anything interesting about certain queries?

- MATCH (p:Person {name: "Keanu Reeves"}) RETURN p.born AS answer
- MATCH (m:Movie {title: "Top Gun"}) RETURN m.released AS release_date
- MATCH (m:Movie {title: "Matrix, The"})< -[:ACTED_IN]-(p) RETURN p.name AS cast
- MATCH (m:Movie)< -[:ACTED_IN]-(p:Person) WHERE m.title = "Matrix, The" RETURN p.name AS names
- MATCH (m:Movie {title: "The Matrix"})< -[:ACTED_IN]-(p) RETURN p.name AS cast
- MATCH (m:Movie {name: "Matrix, The"})< -[:ACTED_IN]-(p) RETURN p.name AS cast

DETACH DELETE 删除关系 + Node.

不能直接删 Node

2. Execute the following query and explore the resulting graph using its edges through the Neo4j Browser UI (e.g. which other movies did the cast act in, etc?).

```
MATCH (m:Movie {title: "Top Gun"}) RETURN m
```

3. What is wrong with the following queries and how could we correct these?

Where 是用 =

- MATCH (p:Person {name="Kiefer Sutherland"}) RETURN p.born AS age → m 不存在 name.
- MATCH (m:Movie)-[:ACTED_IN]->(p:Person) WHERE m.name = "Da Vinci Code, The" RETURN p.name AS cast → m.title.
- CREATE (p:Person) WHERE p.name = "Jane Doe" AND p.born = 2000
create (p:Person {name: " ", born: " "}) No where

4. Execute the following aggregate queries and view the Neo4j Query Planner using "PROFILE".

- MATCH (p:Person)-[:ACTED_IN]->(m:Movie) RETURN m.title AS Movie, COUNT(p) AS Cast_size
返回执行细节
- PROFILE MATCH (p:Person)-[:ACTED_IN]->(m:Movie) RETURN m.title AS Movie, COUNT(p) AS Cast_size
orderby Limit x 数量

5. Execute the following Cypher queries and write down the results.

- MATCH (p:Person) WHERE p.name = "Jane Doe" RETURN p.born
- MATCH (p:Person{born: 2000}) RETURN p.name
- MATCH (p)-[ACTED_IN]->(m) WHERE p.name = "Tom Cruise" RETURN p.title
- MATCH (m:Movie) RETURN m.title, m.released ORDER BY m.title DESC LIMIT 3

6. Write down the following (informal) queries in Cypher.

- (a) List all the movie titles and the respective year they were released ordered (in ascending order) by which year they were released.
- (b) List the names and birth year of all cast members of the movie "Cast Away".

```
Match (m:Movie)
Return m.title and m.released order by m.release ASC

Match (m:Movie{title:"Cast Away"})<-[acted_in]-(a:Actor)
Return a.name , a.born
```

7. Note down and execute the respective Cypher queries to perform the following tasks.

- (c) Create a node with label "Movie" and title "Auckland Adventures", release date as the year 2025 and come up with your own idea of additional properties and respective values (i.e. budget as 250.000, etc)
- (d) Create a relationship / edge between the Person node with name Jane Doe (that we have created earlier) and the Movie node "Auckland Adventures" where the edge points towards the Movie node. Give the edge the label "ACTED_IN" and the property "roles" and for the respective value come up with a list of roles Jane played (e.g. ['Wonder Woman', ...])

8. (Trickier questions) Write down the following (informal) queries in Cypher.

- (e) List the names of all actors that acted in the movie "The Matrix", but not in the sequels of "The Matrix" (you might have to find out the titles of these movies)
- (f) List the name of the oldest actor who starred in "The Da Vinci Code"?

9. Feel free to explore the "Recommendations" graph dataset further and execute queries to practice Cypher.